Report Title: Predicting California Life Expectancy
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GitHub repo: https://github.com/asustayta/MATH-CSCI485_Group_9
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Abstract:
This study investigates how neighborhood-level socio-economic and environmental conditions impact life expectancy in California, utilizing data from the HPI and the CDC. To address the complex, non-linear relationships often overlooked by traditional linear regression models, a comprehensive machine learning pipeline was implemented. Following an 80/20 train-test split, LASSO regression was utilized to identify the 21 most predictive features influencing life expectancy. Two-Way ANOVA analyses confirmed significant geographic disparities in longevity across different neighborhoods, while K-Means clustering categorized these neighborhoods into four distinct socio-economic tiers.
Our research further evaluated advanced ensemble algorithms, specifically Random Forest and XGBoost, both of which achieved a Test R^2 of 0.50. However, an overfitting analysis revealed that the Random Forest model overfitted the training data with a Train R^2 of 0.941, essentially memorizing it. In contrast, XGBoost, which was optimized with early stopping techniques, demonstrated greater robustness and was selected as the superior model.
To interpret the findings, SHAP was applied to the XGBoost model, indicating that education, poverty, and income were the most significant drivers of the community health. Ultimately, the project highlights that a remarkable 50% of the variance in life expectancy can be solely attributed to neighborhood socio-economic conditions, underscoring the critical link between improved community environments and enhanced long-term health outcomes.

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Introduction:
For our project, we want to explore the specific neighborhood factors that impact life expectancy. We decided to use the Healthy Places Index (HPI) dataset because it is incredibly comprehensive. The HPI dataset has the CDC's life expectancy estimates matched up with 23 different community indicators (like education, economics, and environmental factors) for California census tracts.
Community health is significantly shaped by various localized factors, including socioeconomic status, environmental conditions, and demographics. The objective of this project is to forecast life expectancy at birth (LEB) for different census tracts within California. By identifying the

neighborhood-specific indicators, such as the quality of housing, levels of pollution, and educational attainment, that most influence life expectancy, policymakers can strategically direct resources to communities that are most vulnerable. This approach aims to enhance health outcomes by addressing the unique challenges faced by these populations.

Data Collection and Preparation:

We wanted to ensure the highest level of accuracy and data integrity for our models, so we directly contacted the Public Health Alliance of Southern California (PHASoCal) to formally request access to the complete, raw HPI 3.0 data file (2010 geographies).
To construct our model, we used two main datasets: the California Healthy Places Index 3.0 (HPI), which comprises 23 standardized indicators representing various aspects of community health, and the CDC's USALEEP dataset, which provides official life expectancy estimates across the United States. These datasets were chosen to provide a comprehensive foundation for analyzing health disparities and community well-being.

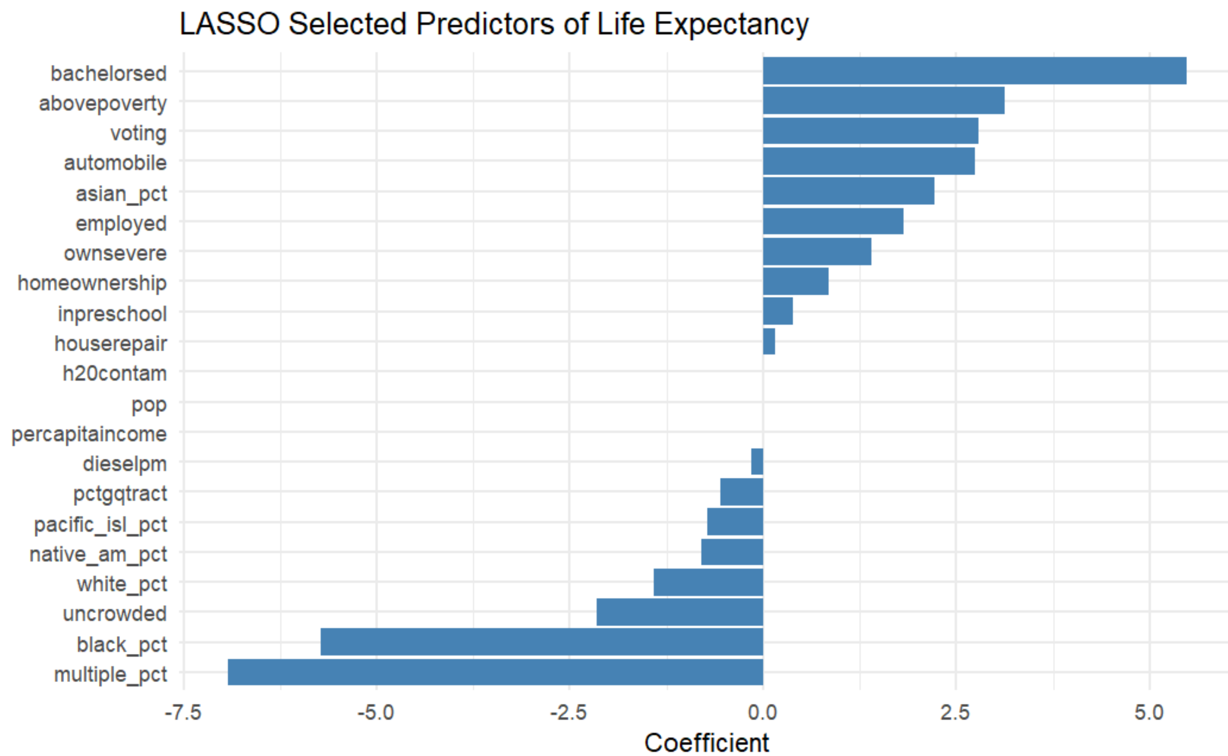
HPI Dataset: We began with the raw HPI 3.0 dataset, containing 8,057 California census tracts and 84 variables. After standardizing column names, removing administrative metadata, and dropping unpopulated tracts, our intermediate dataset was reduced to 8,012 observations and 83 variables. We isolated 79 observations that were missing critical data.

USALEEP Dataset: Our raw CDC life expectancy dataset began with 7,516 observations and 7 variables. We standardized the geographic FIPS IDs and strictly filtered out tracts with suppressed or unreliable life expectancy estimates, resulting in a clean dataset of 7,100 observations and 4 variables.
Final Merged Dataset: By performing a left-join on the 11-digit tract IDs, we successfully merged our clean indicators with the official CDC responses to create our final, analysis-ready dataset.
To evaluate our models, we split the dataset into 80% for training and 20% for testing, using the stratified sampling based on the income quartile. This approach ensured balanced representation across socio-economic tiers, avoiding bias from uneven distributions. A random seed was used for the reproducibility of the data split.

Methodology and Modeling:
We conducted a quantitative analysis in R using some advanced statistical and machine learning techniques. Our modeling consisted of the following:

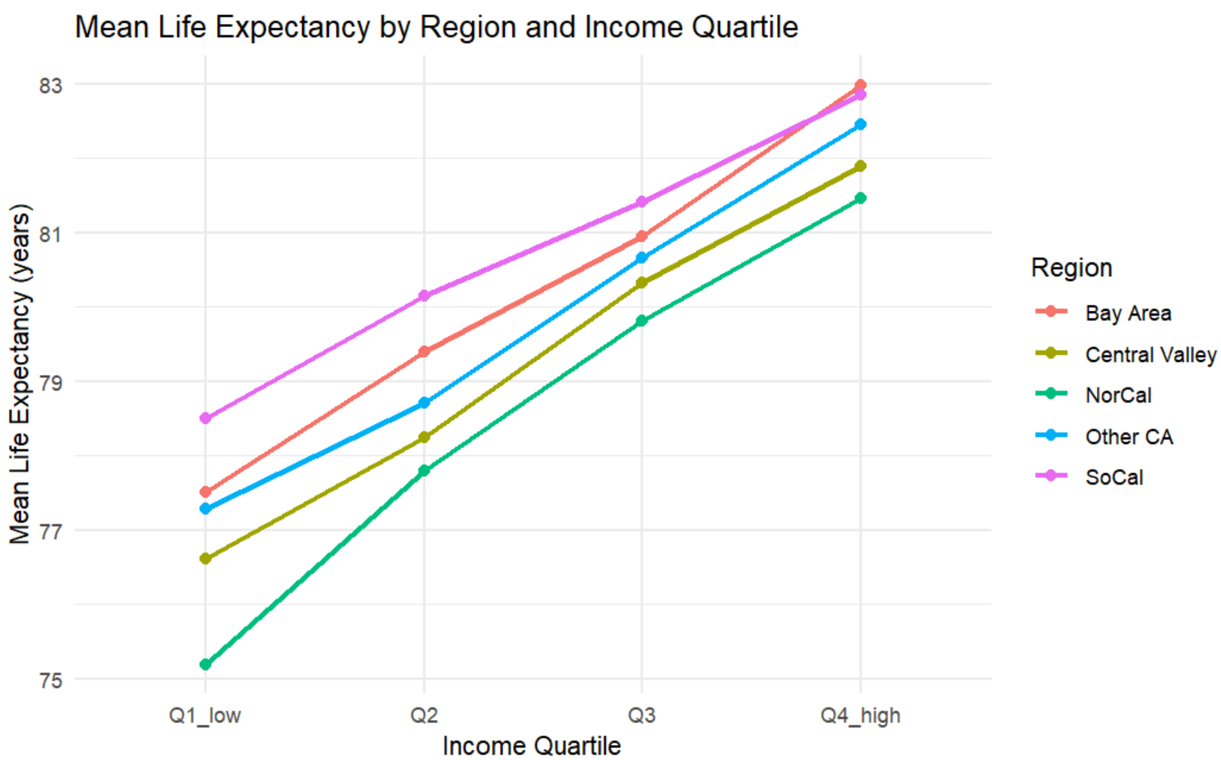
LASSO: We performed a LASSO regression to filter out the noise so that overlapping data wouldn't overwhelm our model. We were left with precisely 21 important predictors after LASSO automatically eliminated the unnecessary variables. This makes it easier to identify the real causes of neighborhood health, showing that things like education level have a greater influence on life expectancy than basic income.

```r
ggplot(lasso_coef_df, aes(x = reorder(variable, coefficient), y = coefficient)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  labs(title = "LASSO Selected Predictors of Life Expectancy",
       x = NULL, y = "Coefficient") +
  theme_minimal()
```



Two-Way ANOVA: We used a two-way ANOVA to evaluate how income quartile and geographic region relate to mean life expectancy. The results indicate a significant effect of income, with higher-income census tracts associated with longer life expectancy. And the *Q-Q Residuals* plot confirms that our data is mostly normally distributed.

```r
anova_df %>%
  group_by(region, income_quartile) %>%
  summarise(mean_leb = mean(leb_cdc, na.rm = TRUE), .groups = "drop") %>%
  ggplot(aes(x = income_quartile, y = mean_leb,
             color = region, group = region)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  labs(title = "Mean Life Expectancy by Region and Income Quartile",
       x = "Income Quartile", y = "Mean Life Expectancy (years)",
       color = "Region") +
  theme_minimal()
```



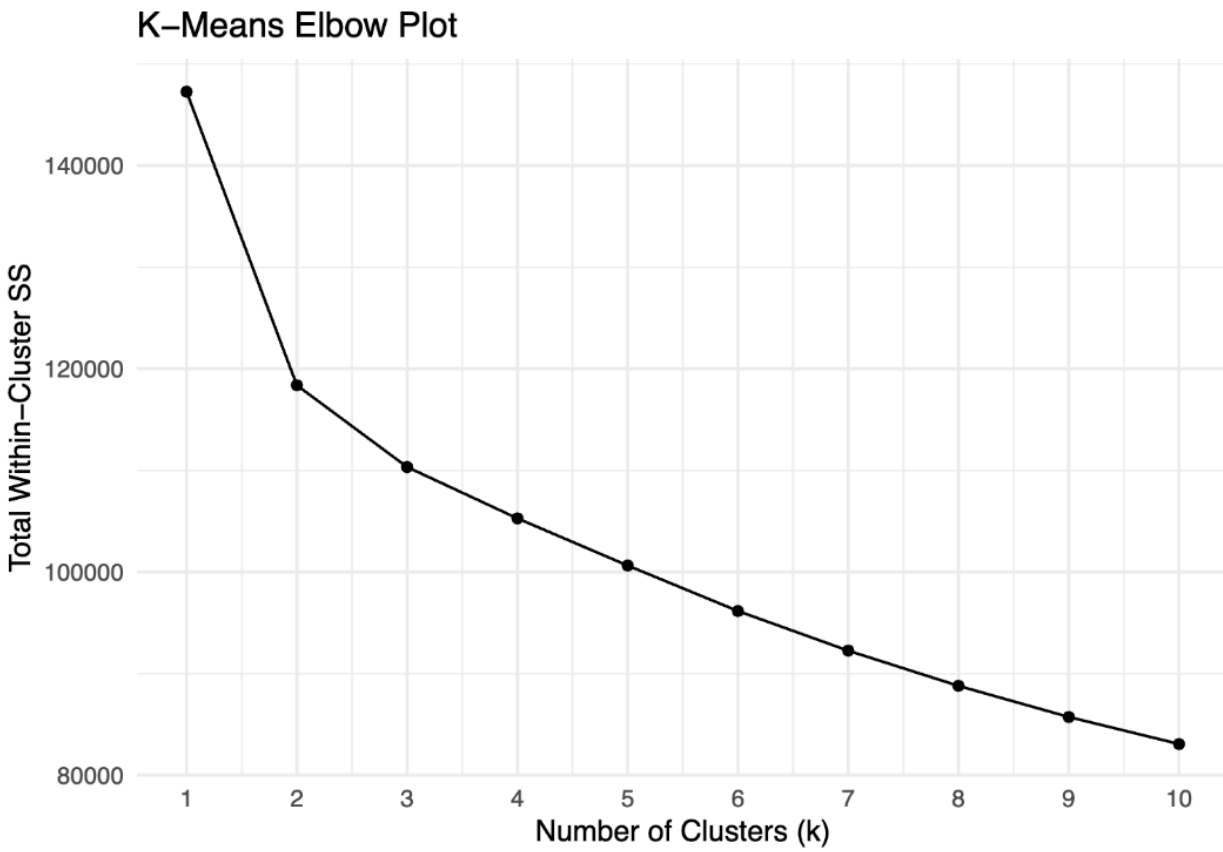
K-MEAN:

We used K-means clustering as an unsupervised learning method to see whether California census tracts naturally grouped together based on the variables that LASSO selected. Since K-means is distance based, all variables had to be scaled before fitting the model. This was important because the predictors are measured on very different scales, such as per capita income, demographic percentages, and environmental indicators.

The goal of K-means was not to directly predict life expectancy, but to better understand whether similar census tracts form meaningful health related groups. We used the LASSO selected variables as the clustering inputs because these variables had already shown some importance for predicting life expectancy. This helped reduce unnecessary noise in the clustering step.

```r
# Use LASSO-selected variables for clustering
cluster_vars <- intersect(selected_vars, colnames(train_data))

cluster_df <- final_df %>%
  select(geo_id, all_of(cluster_vars)) %>%
  drop_na()

# Scale variables (k-means is distance-based, scaling is required)
cluster_scaled <- cluster_df %>%
  select(-geo_id) %>%
  scale()

# Elbow plot — find optimal number of clusters
set.seed(42)
wss <- map_dbl(1:10, ~ {
  kmeans(cluster_scaled, centers = .x, nstart = 25, iter.max = 100)$tot.within
ss
})
```

```r
k_chosen <- 4

set.seed(42)
kmeans_fit <- kmeans(cluster_scaled, centers = k_chosen,
                     nstart = 25, iter.max = 100)
```



The elbow plot was used to compare different values of k. We chose **k = 4** because the decrease in total within cluster sum of squares started to level off around this point. This means that adding more clusters after 4 still improved the clustering, but not by a large amount.

After fitting the model, the clusters had the following sizes:

Cluster	Number of Tracts	Mean Life Expectancy	Mean HPI	Mean Income
4	1,734	78.2	-0.663	18,780
1	2,256	79.5	-0.109	30,282
3	1,066	81.8	0.293	43,041
2	1,958	82.5	0.638	63,459

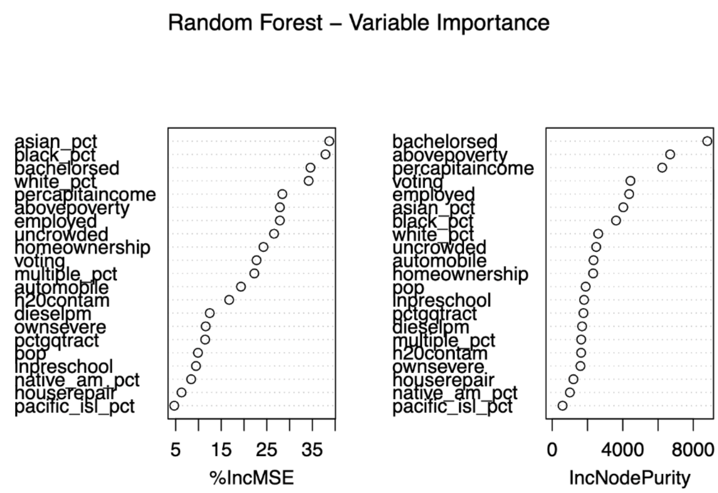
The cluster results show a clear pattern. The cluster with the lowest mean income also had the lowest mean life expectancy and lowest mean HPI score. In contrast, the cluster with the highest mean income had the highest mean life expectancy and highest mean HPI score. This supports the idea that neighborhood level economic and health conditions are connected to life expectancy. The clusters also helped confirm that the variables selected by LASSO were useful because they separated tracts into groups that made sense from a health and socioeconomic perspective.

Random Forest:

We used Random Forest as one of our main predictive models because it can handle nonlinear relationships and interactions between variables. This was useful for our project because life expectancy is likely affected by many factors at the same time, such as education, poverty, income, environment, housing, and demographics.

For Random Forest, we used the same LASSO selected variables as predictors. This allowed us to keep the model more focused and avoid using every possible variable from the HPI dataset. The model was trained with **500 trees**, and the number of variables tried at each split was set to the square root of the number of predictors.

```
rf_vars <- selected_vars

rf_train <- train_data %>%
  select(all_of(c(rf_vars, "leb_cdc"))) %>%
  drop_na()

rf_test <- test_data %>%
  select(all_of(c(rf_vars, "leb_cdc"))) %>%
  drop_na()

set.seed(42)
rf_model <- randomForest(
  x          = rf_train %>% select(-leb_cdc),
  y          = rf_train$leb_cdc,
  ntree      = 500,
  mtry       = floor(sqrt(length(rf_vars))),  # default for regression
  importance = TRUE
)
```



The Random Forest model explained about **51.16%** of the variation in life expectancy during training. On the test set, it had an **RMSE of 2.367** and an **R^2 of 0.500**. This means that the model's predictions were off by about 2.37 years on average, and it explained around half of the variation in life expectancy in the test data.

The variable importance plot showed that several of the most important predictors were related to education, income, poverty, employment, and demographics. Some of the strongest predictors included **bachelorsed**, **abovepoverty**, **percapitaincome**, **voting**, **employed**, **asian_pct**, and **black_pct**. This suggests that life expectancy is not explained by one single factor, but by a combination of socioeconomic and demographic conditions.

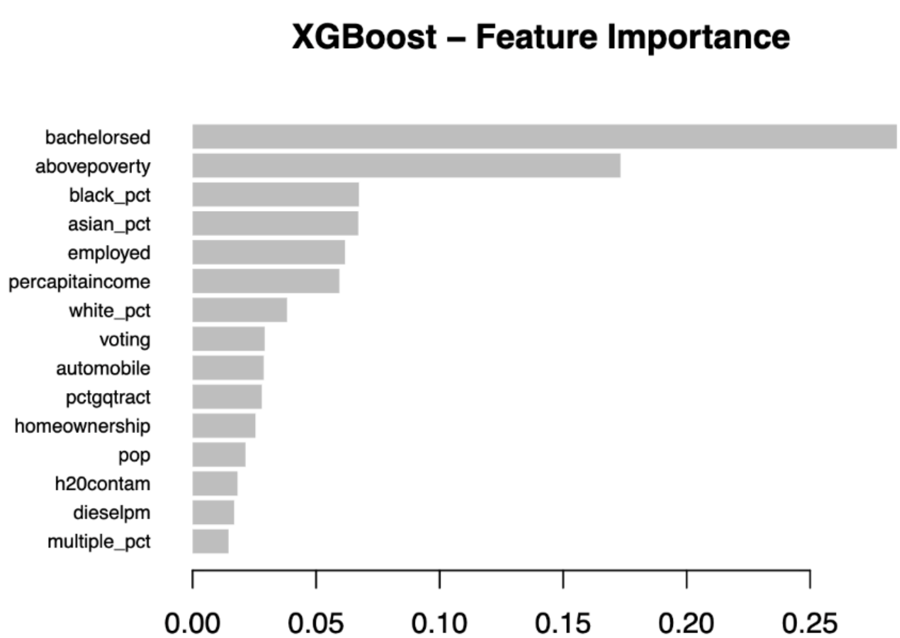
XGBoost:

We also used XGBoost as a second machine learning model. XGBoost is a boosted tree model, meaning it builds many decision trees in sequence, where each new tree tries to improve on the errors of the previous trees. We chose this model because it is often strong for structured tabular data and can capture nonlinear patterns.

Like Random Forest, XGBoost used the LASSO selected variables as predictors. Since XGBoost requires numeric matrices, the predictor data was converted into matrix form before training. We used cross validation to select the best number of boosting rounds. The best number of rounds was **222**.

```
params <- list(
  objective     = "reg:squarederror",
  learning_rate = 0.05,
  max_depth     = 4,
  subsample     = 0.8
)

# Cross-validated XGBoost
set.seed(42)
xgb_cv <- xgb.cv(
  data                  = dtrain,
  nrounds               = 500,
  nfold                 = 10,
  params                = params,
  early_stopping_rounds = 20,
  verbose               = 0
)

best_nrounds <- which.min(xgb_cv$evaluation_log$test_rmse_mean)
```

```
# Fit final XGBoost model
set.seed(42)
xgb_model <- xgb.train(
  data    = dtrain,
  nrounds = best_nrounds,
  params  = params,
  verbose = 0
)
```



The XGBoost model performed slightly better than the Random Forest model. On the test set, XGBoost had an **RMSE of 2.360** and an **R^2 of 0.500**. This was the lowest RMSE out of all the models we tested. The improvement over Random Forest was small, but XGBoost was still our best performing model overall.

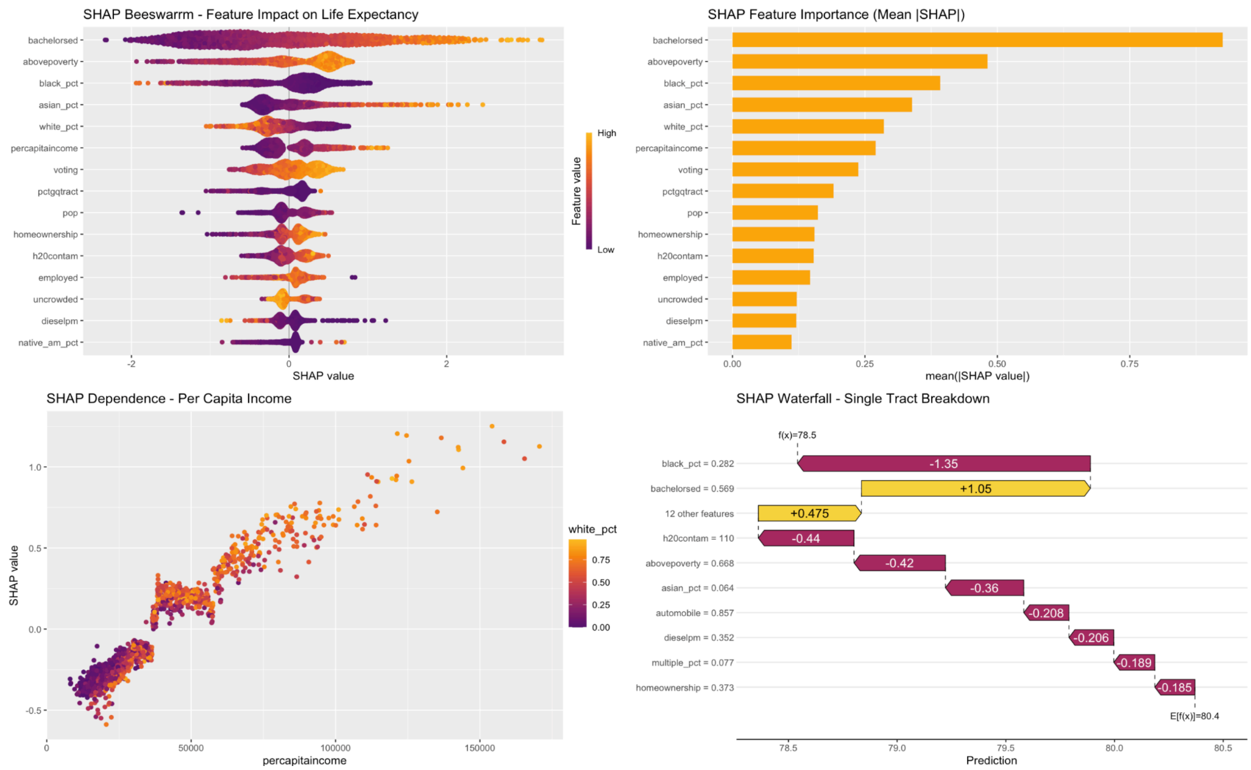
The XGBoost feature importance plot showed that important predictors included **bachelorsed**, **abovepoverty**, **black_pct**, **asian_pct**, **employed**, **percapitaincome**, **white_pct**, **voting**, **automobile**, **pctgqtract**, **homeownership**, **pop**, **h20contam**, **dieselpm**, and **multiple_pct**. These results were similar to the Random Forest model, which gives us more confidence that these variables are consistently related to life expectancy.

SHAP:

To better interpret the XGBoost model, we used SHAP values. SHAP helps explain how each variable contributes to individual predictions and overall model behavior. This was important because tree based models like XGBoost can be harder to interpret than a linear model.

We used SHAP plots to show which variables had the largest impact on predicted life expectancy. The SHAP bar plot showed the average absolute effect of each variable, while the beeswarm plot showed whether high or low values of each variable increased or decreased predictions.

```r
shap <- shapviz(xgb_model, X_pred = xgb_test_x)

sv_importance(shap, kind = "beeswarm") +
  labs(title = "SHAP Beeswarrm - Feature Impact on Life Expectancy")

sv_importance(shap, kind = "bar") +
  labs(title = "SHAP Feature Importance (Mean |SHAP|)")
```



The SHAP results supported the previous model findings. Variables such as **bachelorsed**, **abovepoverty**, **black_pct**, **asian_pct**, **white_pct**, **percapitaincome**, and **voting** had large effects on predicted life expectancy. The SHAP dependence plot for per capita income showed that income generally had a positive relationship with predicted life expectancy, although the relationship was not perfectly linear. This makes sense because income is important, but it does not fully explain health outcomes by itself.

The SHAP waterfall plot was also useful because it showed how the model arrived at a prediction for one specific census tract. For example, some variables pushed the prediction upward while others pushed it downward. This helped show that the model is not only using one variable, but combining many neighborhood conditions to estimate life expectancy.

Results and Analysis

Overall, the models showed that life expectancy in California census tracts can be predicted reasonably well using HPI indicators and demographic variables. The final cleaned dataset contained **7,014 tracts**, with **5,610 training rows** and **1,404 test rows** after the 80/20 train-test split.

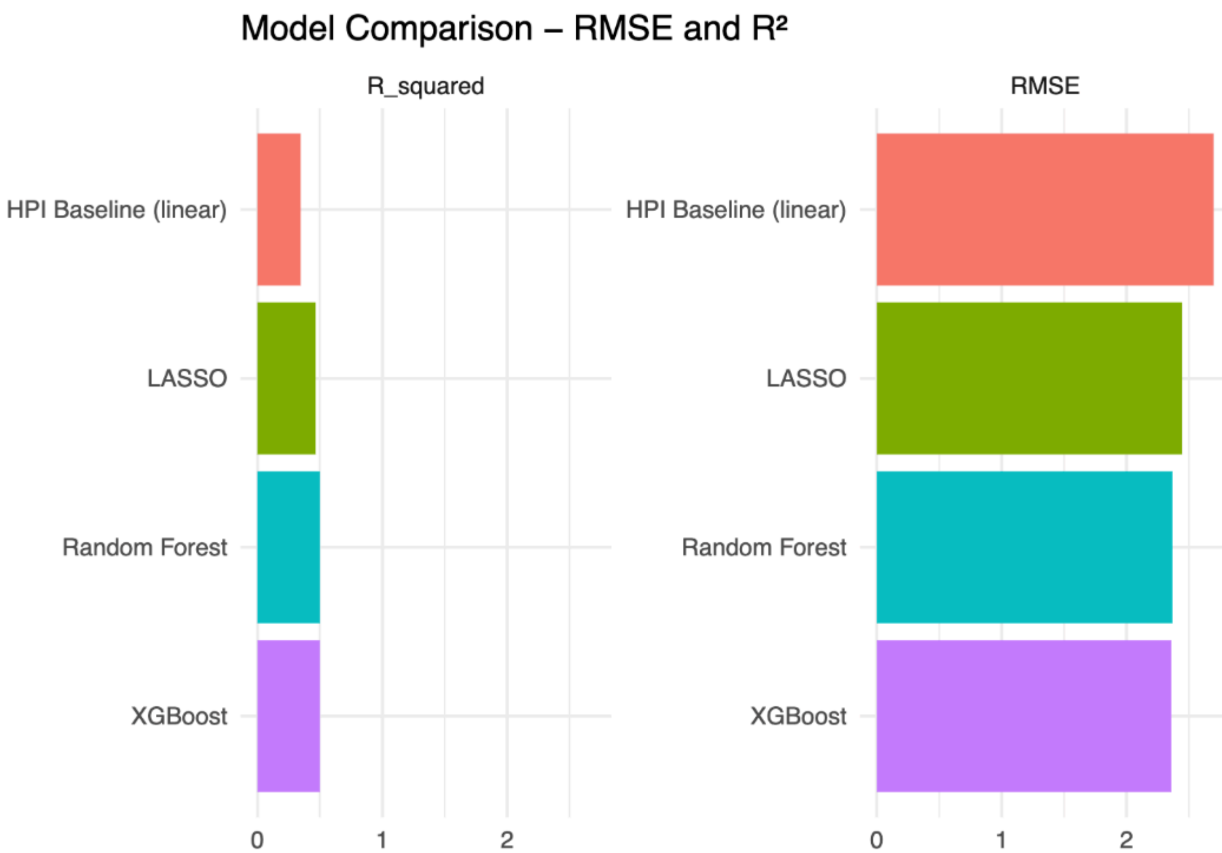
The first important result came from LASSO. LASSO selected **21 variables**, including poverty, education, employment, per capita income, homeownership, environmental variables, and demographic percentages. The LASSO test RMSE was **2.447**, with an **R^2 of 0.464**. This gave us a strong interpretable baseline because it showed which variables had the clearest linear relationship with life expectancy.

The two-way ANOVA showed that both **region** and **income quartile** were statistically significant predictors of mean life expectancy. The interaction between region and income quartile was also significant. This means that the relationship between income and life expectancy is not exactly the same across every region. Income quartile had the strongest effect, with Q4 high income tracts having about **4.01 years higher life expectancy** than Q1 low income tracts based on the Tukey comparison.

The K-means clustering results also supported this pattern. The lowest life expectancy cluster had the lowest mean income and lowest HPI score, while the highest life expectancy cluster had the highest mean income and highest HPI score. This showed that even without using life expectancy directly as the target in clustering, the selected neighborhood variables still grouped tracts in a way that matched health differences.

For the predictive models, XGBoost performed the best, followed very closely by Random Forest. The model comparison was:

Model	RMSE	R^2
XGBoost	2.360	0.500
Random Forest	2.367	0.500
LASSO	2.447	0.464
HPI Baseline Linear Model	2.700	0.346



This comparison shows that both machine learning models improved over the HPI baseline model. The HPI baseline used only the HPI score in a linear model, while our models used multiple selected predictors. XGBoost reduced the RMSE from **2.700** to **2.360**, and improved R^2 from **0.346** to **0.500**. This means the full machine learning approach captured more information than the HPI score alone.

Discussion:

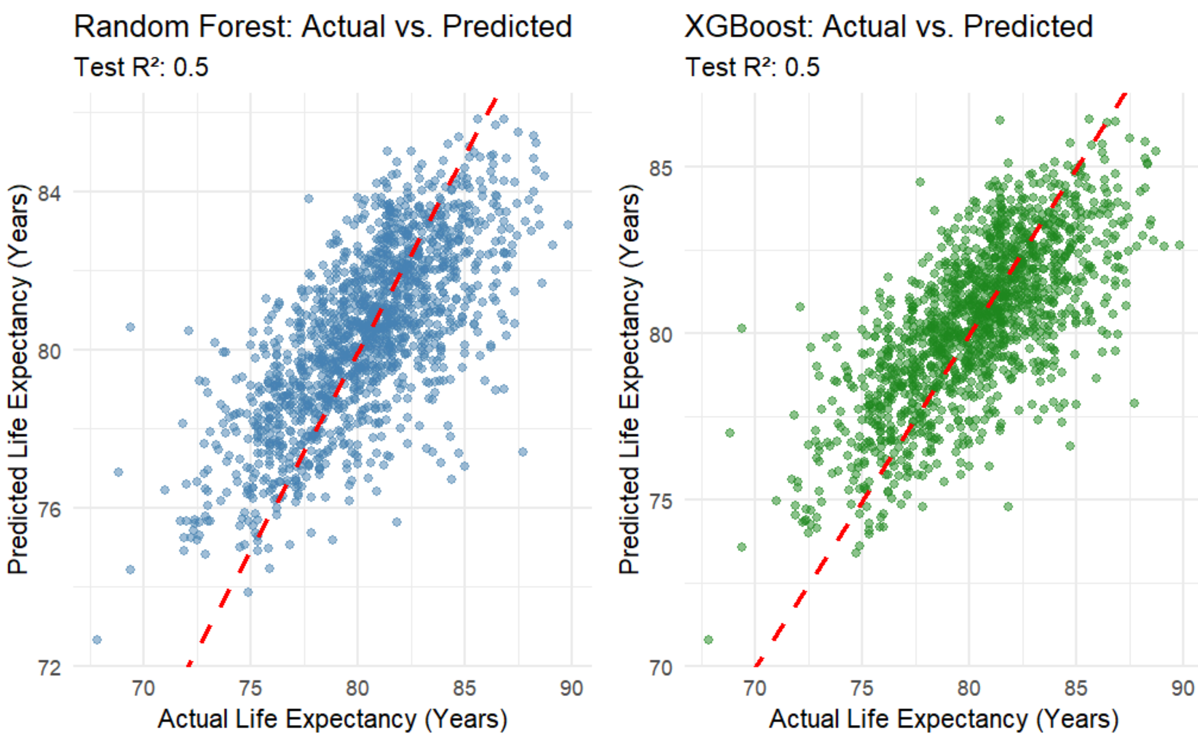
What worked best in this project was using LASSO first to reduce the predictor set, then using those variables in Random Forest and XGBoost. This made the models easier to interpret and helped reduce noise. The SHAP plots also worked well because they made the XGBoost model more understandable.

What did not work as perfectly was prediction accuracy. Even the best model only explained about half of the variation in life expectancy. This suggests that census tract indicators are useful, but they cannot explain everything. Life expectancy is affected by many factors that may not be fully captured in this dataset, such as access to healthcare, individual behavior, local policy, medical history, and other historical or social conditions.

After evaluating our models, it was evident that the advanced models significantly outperformed both the linear HPI Baseline and LASSO models, as demonstrated by reduced error rates (RMSE) and an increase in R^2 values up to 0.5. Despite Random Forest and XGBoost appearing to perform equally in the initial comparison, we investigated potential overfitting. A Train vs. Test analysis was conducted, revealing that Random Forest had memorized the training data, resulting in a significant drop in its performance from 94% accuracy on training data to just 50% on test data. In contrast, XGBoost effectively employed early stopping techniques, allowing it to maintain a stable R^2 of 0.5 on test data, ultimately designating it as the superior model over Random Forest.

```r
# Create the Random Forest Plot
plot_rf <- ggplot(plot_df_rf, aes(x = Actual, y = Predicted)) +
  geom_point(alpha = 0.5, color = "steelblue") +
  geom_abline(intercept = 0, slope = 1, color = "red", linetype = "dashed", linewidth = 1) +
  labs(title = "Random Forest: Actual vs. Predicted",
       subtitle = paste("Test R²:", round(rf_r2, 3)),
       x = "Actual Life Expectancy (Years)",
       y = "Predicted Life Expectancy (Years)") +
  theme_minimal()

# Create the XGBoost Plot
plot_xgb <- ggplot(plot_df_xgb, aes(x = Actual, y = Predicted)) +
  geom_point(alpha = 0.5, color = "forestgreen") +
  geom_abline(intercept = 0, slope = 1, color = "red", linetype = "dashed", linewidth = 1) +
  labs(title = "XGBoost: Actual vs. Predicted",
       subtitle = paste("Test R²:", round(xgb_r2, 3)),
       x = "Actual Life Expectancy (Years)",
       y = "Predicted Life Expectancy (Years)") +
  theme_minimal()

# Combine them side-by-side using the patchwork package
final_comparison_plot <- plot_rf + plot_xgb

# Display the plot
final_comparison_plot
```

```
--- Model Overfitting Check ---
Random Forest - Train R²: 0.941 | Test R²: 0.5
XGBoost       - Train R²: 0.679 | Test R²: 0.5
```



Conclusion and Future Work:

The goal of this project was to predict life expectancy in California census tracts using HPI indicators, demographic variables, and CDC USALEEP life expectancy data. Overall, the project showed that neighborhood level features can explain a meaningful amount of variation in life expectancy. The strongest predictors were generally related to education, poverty, income, employment, housing, and demographic composition.

Our best performing model was XGBoost, with an RMSE of 2.360 and an R^2 of 0.500. Random Forest performed almost the same, while LASSO performed slightly worse but was more interpretable. All three models performed better than the simple HPI baseline model, which had an RMSE of 2.700 and an R^2 of 0.346. This shows that using multiple selected indicators gives a better prediction than only using the overall HPI score. However, after we checked for overfitting, our conclusion slightly changed.

After comparing the Train vs. Test R-squared, we found that Random Forest overfitted the training data (Train R^2: 0.941), while XGBoost, using cross-validation and early stopping, effectively generalized to unseen data, making it the most reliable model.

The K-means clustering results also showed that census tracts naturally grouped into different health and income profiles. The lowest income cluster had the lowest average life expectancy, while the highest income cluster had the highest average life expectancy. This supports the broader conclusion that community conditions are strongly connected to health outcomes.

For future work, an investigation on fairness and model performance across different regions and demographic groups can be done. Since life expectancy is related to inequality, it would be important to check whether the model performs equally well for rural areas, urban areas, low income tracts, and different regions of California. Finally, we could use more recent or additional health data if it becomes available, since the HPI and USALEEP datasets are based on specific time periods and may not fully reflect current conditions.

In conclusion, our project found that machine learning models, specifically XGBoost, can improve life expectancy prediction compared with using the HPI score alone. More importantly, the project showed that education, poverty, income, employment, housing, and demographic factors are all important pieces of understanding community health. This supports the idea that improving neighborhood conditions may also help improve long-term health outcomes.

References:

Centers for Disease Control and Prevention, National Center for Health Statistics. (2018). *U.S.A. Life Expectancy at Birth (USALEEP)* [Data set]. Retrieved from https://www.cdc.gov/nchs/nvss/usaleep/usaleep.html

Public Health Alliance of Southern California. (2022). *The California Healthy Places Index (HPI) 3.0 Complete Data File (2010 Geographies)* [Data set provided upon request]. Retrieved from https://www.healthyplacesindex.org/